Parsa MAJIDI

Embedded Linux & Android BSP Engineer

Embedded Linux and Android BSP Engineer with 2+ years of experience in BSP development, hardware bring-up, and low-level debugging on ARM-based platforms, including NXP i.MX SoCs. Experienced in Linux kernel and device tree customization, bootloader configuration, and hardware/software integration.

M.Sc. thesis focused on ML acceleration using AMD Versal AI Engines, with a strong interest in on-device ML deployment and training.

CORE SKILLS

Embedded Platforms: Embedded Linux, Android AOSP, Yocto Project, BSP Development, ARM-Based Systems
Low-Level Development: Linux Kernel Customization, Device Trees, U-Boot, Hardware Bring-Up, Hardware/Software Integration, Embedded Debugging
Programming: C, C++, Python, VHDL, Verilog
Hardware & Interfaces: I2C, SPI, Bluetooth, CAN, Oscilloscope, Logic Analyzer
Tools: Git, Docker

WORK EXPERIENCE

Kynetics

Embedded Systems Engineer (Aug 2024 – Present)

Main responsibilities:

- Develop and maintain BSPs for Android- and Yocto-based embedded platforms.
- Integrate hardware and software components on ARM-based systems.
- Customize Linux kernels, device trees, and bootloader configurations.
- Debug peripheral drivers and investigate hardware integration issues through schematic analysis.

Key projects and contributions:

- Integrated third-party vendor components, including HALs, proprietary binaries, and VINTF manifests, into Android BSPs.
- Integrated dual-modem Android Radio HAL support on an NXP platform for Quectel EG25-G and Eagle modems, including per-modem VINTF configuration, RIL setup, and SELinux policy.
- Integrated MIPI CSI-2 camera sensors, including AR2020 and AR0246, on the NXP i.MX95 using vendor kernel drivers and device-tree configuration.
- Developed USB Power Delivery firmware in C for a custom iPad stand, improving third-party USB-C hub interoperability from 40% to 90% across tests involving 10 different hub vendors.

OPEN SOURCE & COMMUNITY

- **spi-tools:** Contributed upstream (commit 31c8a46).
- **DNN-FPGA:** Implemented a Keras neural network on an Artix-7 FPGA using HLS4ML, with a custom VHDL UART interface. Project details.
- **ImageUploader:** Developed fixes for a C fork of USBImager, addressing false success reports when writing truncated images and additional pax-archive and backup-related bugs. GitHub repository.

EDUCATION

University of Padova

2025 – M.Sc. in ICT for Internet and Multimedia · Thesis: Tree Tensor Networks on AMD Versal AI Engines
Skills: Vitis, AIE intrinsics and APIs, Versal architecture, quantum-inspired ML (Tree Tensor Networks)

University of Mazandaran

2022 – B.Sc. in Electrical Engineering · GPA: 3.66/4.00

LANGUAGES

Persian Native **English** Professional working proficiency **Italian** Basic

CONTACT

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